

Processor Architecture Report

Enterprise VLSI Chip Design & Verification System

Student: K Murali

Architecture: 32-bit Single-Cycle Processor

1. Introduction

This document describes the architecture of a 32-bit single-cycle processor implemented using Verilog HDL. The processor executes one instruction per clock cycle and demonstrates the complete RTL design flow for educational and internship purposes.

2. Processor Block Diagram

Program Counter → Instruction Memory → Control Unit → Register File → ALU → Data Memory → Write Back

3. Datapath Description

The Program Counter generates the instruction address. The Instruction Memory fetches a 32-bit instruction. The Control Unit decodes the opcode and generates control signals. The Register File supplies operands to the ALU. The ALU performs arithmetic or logical operations. For LOAD and STORE instructions, the Data Memory is accessed. Finally, the write-back path updates the destination register.

4. Instruction Format

Bits	Field
31:28	Opcode
27:25	Destination Register (Rd)
24:22	Source Register 1 (Rs1)
21:19	Source Register 2 (Rs2)
18:0	Immediate/Data Field

5. Supported Instructions

Instruction	Function
MOVI	Load Immediate
ADD	Addition
SUB	Subtraction
AND	Bitwise AND
OR	Bitwise OR
LOAD	Read data from memory
STORE	Write data to memory
NOP	No operation

6. Execution Flow

Fetch → Decode → Execute → Memory Access → Write Back. The processor completes these stages within a single clock cycle.

7. Module Summary

Program Counter: Generates instruction addresses.

Instruction Memory: Stores program instructions.

Control Unit: Generates control signals from the opcode.

Register File: Stores eight 32-bit registers.

ALU: Executes arithmetic and logical operations.

Data Memory: Supports LOAD and STORE instructions.

Processor: Integrates all modules into a complete CPU.

8. Design Workflow

Specification → RTL Design → Functional Simulation (ModelSim) → Quartus Synthesis → RTL Viewer → Technology Map → Timing Analysis → Documentation

9. Applications

FPGA prototyping, processor architecture education, digital logic laboratories, RTL verification training, and embedded system research.

10. Conclusion

The proposed architecture demonstrates a modular, scalable, and synthesizable 32-bit processor suitable for RTL design, functional verification, and FPGA implementation.